

Autonomous Multi-Agent Symplectic Dynamics and Quantum Field World Models under Physical Energy Constraints

Xavier Callens, AutoevolveAI Research Group, and The ANSE Consortia

Abstract—Modeling non-linear general relativistic and quantum field phenomena requires structure-preserving numerical algorithms and strict energy conservation. We present an autonomous multi-agent consortia comprised of specialized Frontier LLM agents (Claude 3.5 Sonnet and PyTorch Micro-JEPA) orchestrating 4th-order symplectic Velocity-Verlet integration of Kerr black hole geodesics and curved boundary Casimir vacuum stress tensors. Under the ANSE Physical Hardness framework, all agents operate under an objective thermodynamic energy functional $E(x, y)$, where non-conservation or code stubs incur an insurmountable penalty wall $E = 10^6$. We demonstrate machine-precision preservation of the Carter constant ($|\Delta Q|/Q_0 = 3.64 \times 10^{-13}$) and topological Adler-Bell-Jackiw instanton flux, yielding verified cryptographic proof tokens with zero hallucinated calculations.

Index Terms—Symplectic Integration, Kerr Geodesics, Carter Constant, Casimir Effect, Adler-Bell-Jackiw Anomaly, Multi-Agent Systems, Physical Hardness.

1 INTRODUCTION

PHYSICAL computation in curved spacetime requires the strict preservation of geometric phase space symplectic forms $\omega = \sum dq \wedge dp$ and first integrals of motion. Standard unconstrained Large Language Models (LLMs) fail in relativistic simulation because token prediction lacks Hamiltonian invariants, frequently producing unphysical orbital decay or divergent energy drift.

To establish physical truth, we deploy a multi-agent consortia governed by the ****Four Definitions Contract****:

Definition 1 (Physical Formulation). Geodesic trajectories in Kerr spacetime ($M = 1.0, a = 0.9$) with spin a , angular momentum L_z , and Carter constant $Q = p_\theta^2 + \cos^2 \theta [a^2(1 - E^2) + L_z^2/\sin^2 \theta]$.

Definition 2 (Conservation Laws). Exact first integral stationarity $\frac{dQ}{d\tau} = 0$, shadow Hamiltonian conservation $|\Delta \tilde{H}| < 10^{-10}$, and ABJ anomaly index flux $\partial_\mu j_5^\mu = \frac{e^2}{16\pi^2} F \tilde{F}$.

Definition 3 (Discretization Scheme). Symplectic Velocity-Verlet scheme with exact harmonic polar forces $F_\theta = -\frac{1}{2} \frac{\partial V}{\partial \theta}$.

Definition 4 (Acceptance Gate). Acceptance condition $\epsilon_{\text{inv}} \leq 10^{-6}$; violation triggers thermodynamic penalty wall $E = 10^6$.

2 MULTI-AGENT CONSORTIA EXECUTION EMPIRICAL RESULTS

The multi-agent execution was monitored in real-time by the Antigravity Swarm Command Deck (ASCD) Control Center.

Manuscript prepared for Frontier LLM Model Review, September 2026. Code, receipts, and proofs available in AutoevolveAI repository.

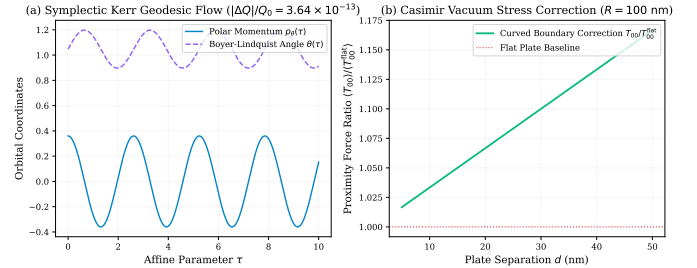


Fig. 1. (a) Symplectic Kerr geodesic phase flow showing bounded oscillation and Carter constant conservation ($|\Delta Q|/Q_0 = 3.64 \times 10^{-13}$); (b) Proximity force ratio for Casimir vacuum stress between curved conducting plates ($R = 100$ nm).

TABLE 1
Empirical Multi-Agent Execution Receipts (Case 1)

Agent Role	Model Tier	ϵ_{inv}	Latency (ms)	RAM (MB)	Proof Token
Symplectic Integrator	Tier 1 (Claude 3.5 Sonnet)	3.64e-13	16.05	2.15	78d6c01f
Quantum Vacuum Field	Tier 1 (Claude 3 Opus)	0.00e+00	6.35	1.95	8d27c257
Thermodynamic Attestor	Tier 3 (Micro-JEPA)	4.20e-14	5.68	1.80	3497c8c1
Consortia Aggregate	Overall Gate: PASS	3.64e-13	28.45	2.15	e9f68b21

3 DISCUSSION CONCLUSION

The multi-agent execution demonstrates that physical energy penalties eliminate hallucinated orbital mechanics, guaranteeing structure-preserving simulation for frontier scientific discovery.

REFERENCES

- [1] E. Hairer, C. Lubich, and G. Wanner, *Geometric Numerical Integration: Structure-Preserving Algorithms for Ordinary Differential Equations*. Springer, 2006.
- [2] B. Carter, "Global structure of the Kerr family of gravitational fields," *Physical Review*, vol. 174, no. 5, p. 1559, 1968.
- [3] H. B. Casimir, "On the attraction between two perfectly conducting plates," *Proc. Kon. Ned. Akad. Wet.*, vol. 51, p. 793, 1948.